🔑 **Answer Key — Teacher Copy · Lesson 6-4**

MSTAR PRACTICE WORKSHEET

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Grade 6 Mathematics · Standard 6.AT.6c

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Evaluate $3 + 4 \times 2$.

A. 9

B. 11

C. 14

D. 24

Multiplication comes before addition: $4 \times 2 = 8$, then $3 + 8 = 11$.

Question 2 SELECTED RESPONSE — CORRECT: A

What does 2^3 mean?

A. $2 \times 2 \times 2$

B. 2×3

C. $2 + 2 + 2$

D. 3×3

The exponent says how many times the base is used as a factor, so $2^3 = 2 \times 2 \times 2 = 8$.

Question 3 SELECTED RESPONSE — CORRECT: A

Sofia buys 4 tickets at $20 each and 2 tickets at $15 each. Which expression shows the total?

A. $4 \times 20 + 2 \times 15$

B. $4 + 20 + 2 + 15$

C. $(4 + 2) \times (20 + 15)$

D. $4 \times 2 + 20 \times 15$

Each ticket type is a count times a price, and the two costs are added: $4 \times 20 + 2 \times 15$.

Question 4 SELECTED RESPONSE — CORRECT: C

Evaluate $5^2 + 3$.

A. 13

B. 16

C. 28

D. 64

Powers come first: $5^2 = 25$. Then add: $25 + 3 = 28$.

Question 5 SELECTED RESPONSE — CORRECT: B

What is the total of $2 \times 15 + 4 \times 20 + 8 \times 2^3$?

A. 126

B. 174

C. 210

D. 1,392

Power first: $2^3 = 8$. Multiply: 30, 80, 64. Add: $30 + 80 + 64 = 174$.

Question 6 SELECTED RESPONSE — CORRECT: D

Miguel evaluates $12 - 2 \times 3$ and gets 30. Margaret gets 6. Who is right?

A. Miguel, because he worked left to right

B. Both are right

C. Neither is right

D. Margaret, because multiplication comes before subtraction

Multiplying first gives $2 \times 3 = 6$, then $12 - 6 = 6$. Miguel subtracted first, getting $10 \times 3 = 30$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Evaluate the numerical expression $3 \times 4^2 - 8$.

A. 16

B. 24

C. 40

D. 136

Following the order of operations, evaluate the power first: $4^2 = 16$. Then multiply: $3 \times 16 = 48$. Then subtract: $48 - 8 = 40$.

- **A:** Only the power was evaluated. Continue: multiply by 3, then subtract 8.
- **B:** The exponent was treated as a multiplier, 4 times 2. Four squared is 16.
- **D:** The multiplication happened before the power. Powers come first: evaluate 4 squared, then multiply.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

A. Multiply 3×4 first, then square the product: $12^2 = 144$, and $144 - 8 = 136$.

B. Evaluate the power first: $4^2 = 16$. Then multiply: $3 \times 16 = 48$. Then subtract: $48 - 8 = 40$.

C. Read 4^2 as $4 \times 2 = 8$, then multiply $3 \times 8 = 24$ and subtract: $24 - 8 = 16$.

D. Subtract first because 16 and 8 are next to each other: $16 - 8 = 8$, then $3 \times 8 = 24$.

The order of operations puts exponents before multiplication, and multiplication before subtraction. Each other path breaks that order.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Subtracting Too Soon

> Devon evaluated $20 - 2 \times 3^2$. His work was: $3^2 = 9$, then $20 - 2 = 18$, then $18 \times 9 = 162$. He wrote that the value of the expression is 162.

Explain Devon's misconception and give the correct value of the expression.

Model answer: Devon evaluated the power correctly but then subtracted before multiplying. In PEMDAS, multiplication comes before subtraction: $3^2 = 9$, then $2 \times 9 = 18$, then $20 - 18 = 2$. The value is 2.

2 points	Student explains that the order of operations requires multiplying 2×9 before subtracting from 20, and gives the value 2.
1 point	Student gives the value 2 but does not explain that multiplication comes before subtraction.
0 points	Student agrees with Devon that the value is 162 or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Seven friends split the bill for a pizza party. They order 4 pizzas at 3^2 dollars each, plus a one-time $6 delivery fee. Write ONE numerical expression for what each friend owes, evaluate it, and explain why the grouping symbols matter.

Model answer: Each friend owes $(4 \times 3^2 + 6) \div 7$. Evaluating: the power first, $3^2 = 9$; then multiply, $4 \times 9 = 36$; then add inside the grouping symbols, $36 + 6 = 42$; then divide, $42 \div 7 = \$6$ each. The grouping symbols matter because without them the expression would read $4 \times 3^2 + 6 \div 7$, which divides only the delivery fee among the friends and leaves each of them paying for all four pizzas — about $36.86 instead of $6. The parentheses are what say "split the WHOLE bill."

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.